COA

Assignment - 4

Section - A

Ans1. SRAM vs DRAM → SRAM uses flip flops to store data, making it faster, expensive & power consuming. It is used in cache memory.
DRAM uses capacitors to store data, needs refreshing, slower but cheaper. It is uses as main memory.

Ans 2. Programs tend to access a small portion of memory repeatedly
Type
→ Temporal locality - recently accessed data is reused
→ Spatial locality → nearby locations are accessed.

Ans 3. Cache memory stores frequently used data/instructions closer to CPU. It reduces avg memory access time & improves performance.

Ans 4. Cache Indexing

- Direct mapped cache
- Set associative cache
- fully associative cache

Ans-5 Larger cache block size increases spatial locality (fewer misses) but increases miss penalty & may load unnecessary data.

Ans-6 Associativity is the number of locations where a memory block can be placed in cache. Higher associativity reduces conflict misses but increases H/W cost.

Ans-7 Large cache size reduces miss rate & improves performance but increases cost, power & access latency.

Ans 8. (i) Write-through: Data is written to both Cache & main memory at same time.

(ii) Write-Back: Data is written to cache first & main memory is updated only when the block is replaced

Ans 9 AMAT = Hit time + (Miss rate × Miss Penalty)

time to access cache — Probability of miss — Time to fetch block from main memory.

Ans 10: LRU block is replaced first. The block which has not been used for longest time is removed

Ans 11: Memory is divided into multiple modules & accessed simultaneously in round robin fashion It increases memory bandwidth & improves performance

Ans-12 Tracks are concentric circular paths on disk surface. Sectors are sub divisions of tracks used to store data blocks.

Ans-13 Flash Memory is a non-volatile memory that retains data even power is off. Used in SSD's, pen drives, memory cards. Faster than magnetic disks & has no moving parts.

Ans-14 I/O Devices share the same address space as memory. Devices are accessed using normal delivery instructions

Ans-15 I/O devices have a separate address space. They are accessed using special I/O instructions

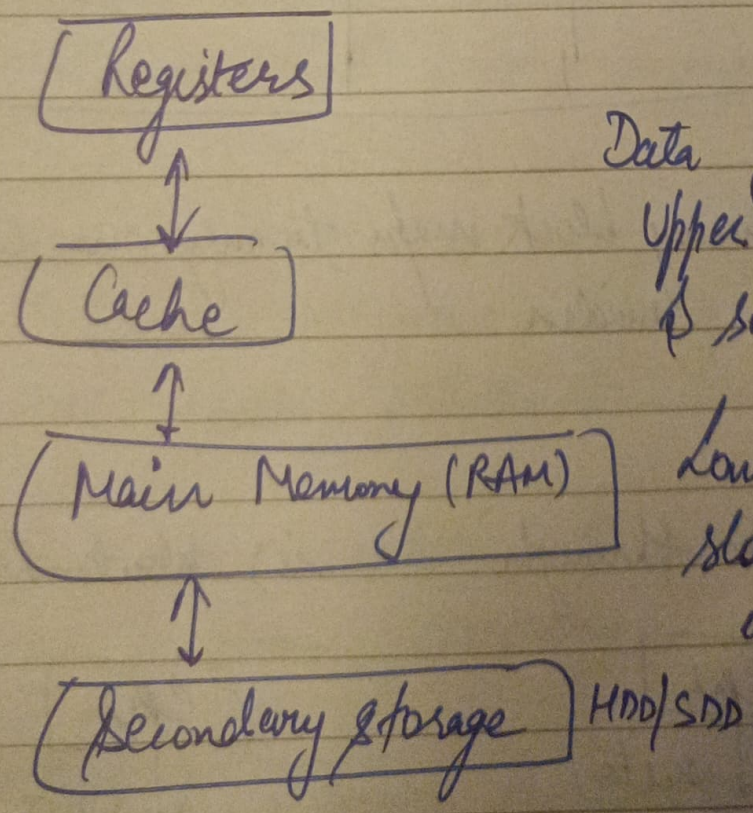
Ans-16 CPU continuously checks the status of I/O devices & controls the data transfer. CPU is busy waiting.

Ans-17 Device interrupts the CPU when its ready. CPU can perform other task meanwhile. Better CPU utilization

Ans-18 DMA allows I/O devices to transfer data directly to user from memory without CPU involvement. It increases efficiency.

Section-B

Ans1



Data flows b/w diff levels. Upper levels are faster & smaller.

Lower levels are slower but larger in capacity.

Ans 2.

Address		
Tag (t bits)	Index (i bits)	offset (b bits)

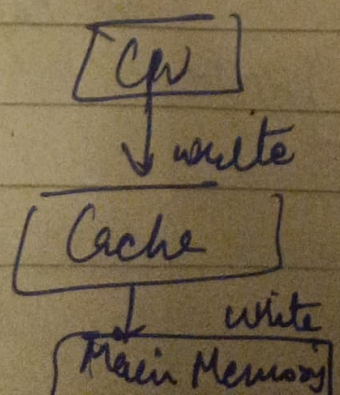
Cache levels

	Valid	Tag	Data block (2^b byte)
0			
1			
2			
$2^i - 1$			

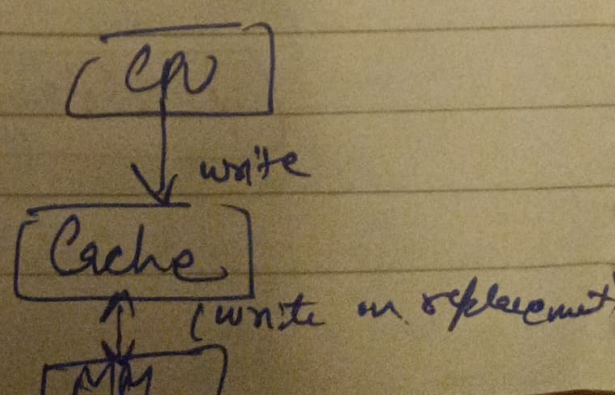
Each memory block maps to only one cache line using index

Ans-3.

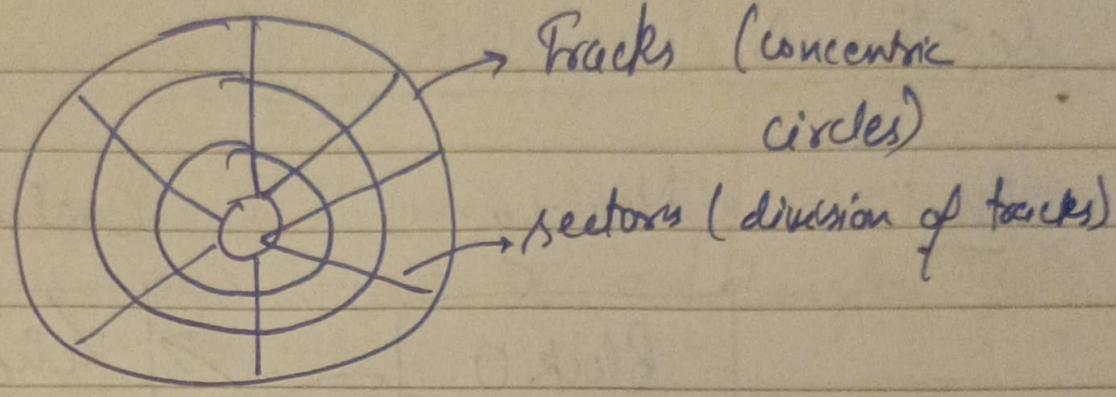
(i) Write through



ii) Write-Back



Ans-4



Section-C

Ans-1 Temporal locality, because the same location is accessed again & again.

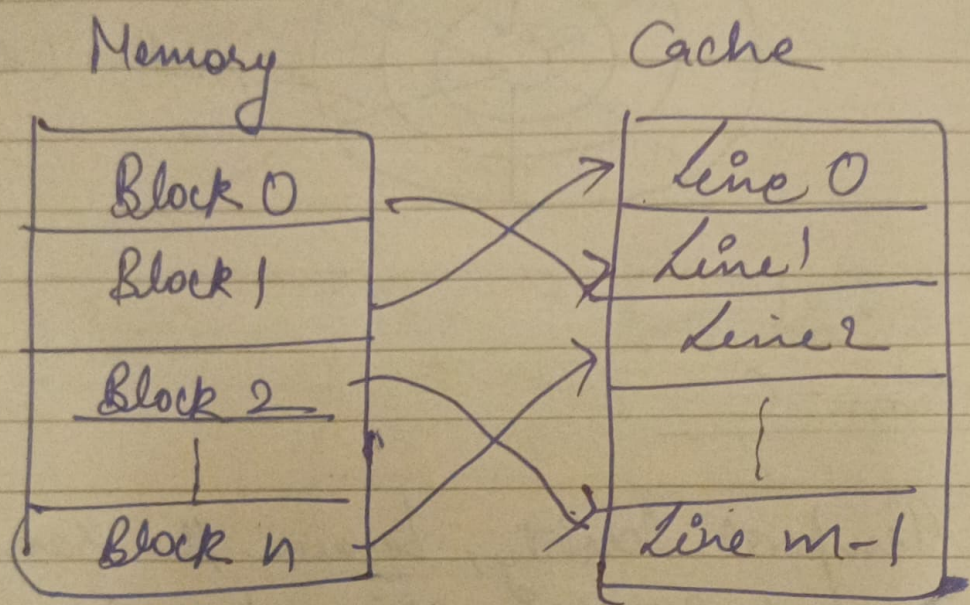
Ans-2 Larger block increase miss penalty (more data to be transferred) & may load unnecessary data (cache pollution)

Ans-3 On every write, data is written to both cache & main memory.

Ans-4 Direct Memory Access (DMA)

Section-D

Ans-1



* Mapping: memory block is placed in cache line using index. Tag is used in index correctness.

Ans-2 $\boxed{\text{AMAT} = \text{Hit time} + (\text{Miss rate} \times \text{Miss Penalty})}$

Factors affecting Performance
- → Lower hit time
- → Lower miss rate
- → Lower miss penalty

These reduces AMAT & improve performance.

Ans 3

Programmed I/O	Interrupt I/O	DMA
CPU checks device continuously	Device interrupts CPU when ready	DMA controller transfers data
CPU is busy waiting	Better CPU utilization	Best efficiency

Section-E

Ans-1 Two assumptions while explaining cache operation
- ↳ fixed block size is assumed
- ↳ Single level cache is considered.

Ans 2 Increases memory traffic because every write updates main memory

Ans 3 Cache Mapping is more diff because it involves address breakdown and mapping logic which is conceptual & needs practi